Prince Justice Sena

AI Engineer · Agentic Payments

justicessiel@gmail.com justiceessielp.github.io github.com/JUSTICEESSIELP linkedin.com/in/essiel-justice x.com/sena_justice

Intelligence is becoming something anyone can hold. I build the agents that turn it into real work — and the rails that let them pay their own way. Built Solvador, winner of Coinbase Batches 003.

WORK EXPERIENCE

Founder & AI Product Engineer — Workplacefiles AI 2026 — Now

Enterprise AI agents for a team's recurring work, on Mac and Windows. Team files, Slack and Google Drive are organised as Markdown and indexed into a knowledge graph; agents retrieve context with GraphRAG backed by vector retrieval, plus a memory layer that carries team decisions between runs. Updates are proposed as reviewable changes with full version history; agents flag missing information instead of guessing.

Co-founder & Lead Developer — Shika Protocol · OffGrid Labs 2025 — 2026

Modular fintech stack on Base for African markets — identity, cedi stablecoin, ROSCA savings, lending and USSD access. 3,000+ users, about 430 new users a month, 85%+ loan repayment.

Co-founder & CTO — EcoFundMe · EcoClime Africa 2024 — Now

Climate crowdfunding with milestone-gated escrow — funds in USDC on Base are released only as verified project milestones are met. Won a $50,000 Avail grant; Renew Capital SPRINT cohort.

Core Developer — The Graph Protocol · Remote 2023 — Now

Build Web3 indexing infrastructure and blockchain data-querying systems used by developers worldwide; contribute to protocol upgrades and smart contract work for DeFi apps.

Developer Relations — DeFi Africa · Remote 2023 — 2024

Grew the African Web3 developer community through tutorials, AMAs with DeFi founders and workshops on DeFi adoption.

CTO — Blockdor 2023

Blockchain mine management for rubber mines in Côte d'Ivoire and gold mines in Ghana, with IoT fingerprint locks.

Machine Learning Engineer — Sidakye Solution Limited · Accra 2021 — 2023

Built "Dave", an AI analytics platform for Ghanaian institutional investors — GDP-growth prediction models ($R^2 \geq 0.6$), macroeconomic and news-sentiment data pipelines, and a REST API serving predictions.

AI PROJECTS

Agentic AI & payments era · 2025 — Now

Solvador — solvador.com 2026

The payment layer agents use to pay for APIs: an x402 facilitator that verifies and settles USDC payments on-chain across 15 mainnets, so servers never hold keys or pay gas. Exact, metered (upto) and batch-settlement schemes; confidential settlement on Base hides which merchant was paid. Winner of Coinbase Batches 003; listed in the official x402 facilitator directory.

AgentSound Demo

Pay-per-play music built on Solvador. Ask an AI agent for a song, and every play pays the artist about a cent from a prepaid balance within a weekly budget. Artists are paid privately in batches on Base and Arbitrum.

GPT & LLM era · 2023 — 2024

KidFriendlySocial 2023

A safe social media web app for kids — BERT detects bad language and GPT-3 recommends content. Deployed with Docker and Kubernetes on DigitalOcean, with CI/CD and monitoring.

Pre-GPT · Deep learning · 2020 — 2022

Language Identification — learning project 2020

Rebuilt the Hasso Plattner Institute's CRNN language-identification model (ICONIP 2017) to learn deep learning — a convolutional + recurrent network that tells apart six spoken languages, served through a small web app.

BLOCKCHAIN & ZK PROJECTS

ZK Rollup L2 Governance — zero-knowledge layer 2 for on-chain governance and voting; Go backend, Avail data availability. Won a $50K Avail grant. 2025

Offline Uniswap DEX — offline version of the Uniswap protocol. 1st place, Arbitrum × StackUp. 2024

Gyema — peer-to-peer delivery using existing travel routes. Winner, Ethereum Safari Hackathon. 2024

Offline Credit Scoring — credit scoring for emerging markets that works offline. Winner, Stellar Surge. 2024

AWARDS

Coinbase Batches 003 — winner with Solvador

Avail Grant — $50,000 — for EcoFundMe and the ZK Rollup L2 Governance 2025

QuickNode Accelerator — with EcoFundMe; pitched to Dragonfly, Halo Capital and other VCs 2025

Arbitrum × StackUp Hackathon — 1st Place — $2,500 and a sponsored trip to DevCon 2024 2024
Ethereum Safari Hackathon — Winner — Gyema 2024
Stellar Surge Pitch Competition — Winner — offline credit scoring 2024
Renew Capital SPRINT — selected with EcoFundMe 2024
Ethereum Accra Hackathon — 1st Place — EcoFundMe prototype 2023
5+ more hackathon awards — blockchain and fintech 2022 — 2024

LEADERSHIP

Graph Africa — co-built the community and onboarded 200+ African builders
Africans Who Invest — Fellow & Facilitator — investment education and mentoring for students 2023 — Now
Tech Lead — 2024 Parliamentary Campaign — offline voter-registration data tool for a parliamentary candidate in Ashaiman Municipality; led planning and the team 2024
Assistant Director & STEM Instructor — Kingscourt International School — taught maths and science; started a Python club 2018 — 2020

TALKS & PITCHES

QuickNode Accelerator Pitch Day — pitched EcoFundMe to Dragonfly, Halo Capital and 7 other VCs 2025
UNICEF Digital Public Goods Stakeholder Forum — presented EcoClime Africa 2024
Stellar Surge Pitch Competition — offline credit scoring; won 2024

WRITING

Zero Knowledge Proofs Series — six-part series on dev.to covering finite fields, modular arithmetic, groups and rings, discrete logarithms, key exchange and elliptic curves (2024). dev.to/justiceessielp

CERTIFICATIONS

ZK Bootcamp — Encode Club (on-chain certificate) Q1 2024
Accounting Mastery · **Valuation for Startups** — Financeable 2024
Product Management in Artificial Intelligence 2022
Blockchain Development Mastery — Polygon 2022

SKILLS

AI & LLMs: LLM agents, agent memory, GraphRAG, vector RAG, human-in-the-loop review, voice agents, agentic payments (x402), Claude, xAI Grok, GPT-3, BERT

Languages: Python, Go, Solidity, TypeScript, JavaScript, Rust, SQL

Blockchain: smart contracts, ERC-20, EIP-3643, DeFi, The Graph, Ethereum, Arbitrum, Base, Stellar, Foundry, x402

Zero knowledge: Halo2, Circom, Noir

Machine learning: TensorFlow, scikit-learn, Pandas, NumPy, NLP, PCA, statistical modelling

Fintech: mobile money, USSD, credit scoring, P2P lending, stablecoins, CBDC design, KYC/AML

Tools: Git, Docker, Kubernetes, GraphQL, Node.js, REST APIs, AWS/GCP

EDUCATION

BSc Earth Science — University of Ghana 2018 — 2022